

Linear Regression

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Outline

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- 5 Polynomial Regression

Data as Vectors and Matrices

Observations, features and labels

i	features			label
	TV	radio	newspaper	sales
1	230.1	37.8	69.2	22.1
2	44.5	39.3	45.1	10.4
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
200	232.1	8.6	8.7	13.4

- Each row is one **observation** (a market): $n = 200$ observations
- The columns TV, radio, newspaper are the **features** (predictors, inputs, independent variables): $p = 3$, budgets in thousands of dollars
- The column sales is the **label** (response, output, target, dependent variable): what we want to predict, in thousands of units

Vectors and matrices

- Observation i is a **feature vector** $\mathbf{x}^{(i)} = (x_1^{(i)}, x_2^{(i)}, \dots, x_p^{(i)})^\top \in \mathbb{R}^p$ with label $y^{(i)} \in \mathbb{R}$; for example $\mathbf{x}^{(1)} = (230.1, 37.8, 69.2)^\top$ and $y^{(1)} = 22.1$
- Stacking the n observations gives the **feature matrix** $\mathbf{X} \in \mathbb{R}^{n \times p}$, whose row i is $(\mathbf{x}^{(i)})^\top$, and the **label vector** $\mathbf{y} \in \mathbb{R}^n$

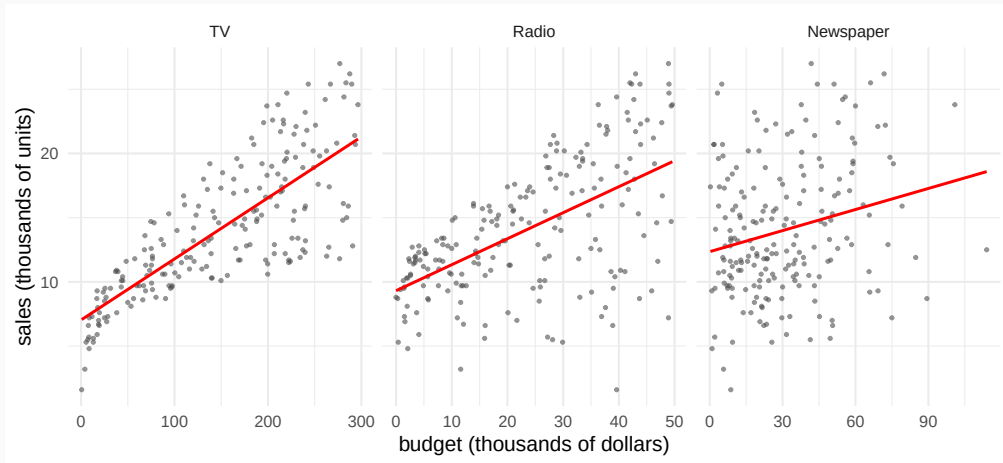
$$\mathbf{X} = \begin{bmatrix} 230.1 & 37.8 & 69.2 \\ 44.5 & 39.3 & 45.1 \\ \vdots & \vdots & \vdots \\ 232.1 & 8.6 & 8.7 \end{bmatrix} = \begin{bmatrix} (\mathbf{x}^{(1)})^\top \\ (\mathbf{x}^{(2)})^\top \\ \vdots \\ (\mathbf{x}^{(n)})^\top \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 22.1 \\ 10.4 \\ \vdots \\ 13.4 \end{bmatrix}$$

- Superscript (i) = which observation (row), subscript j = which feature (column): $x_j^{(i)}$ is entry (i, j) of $\mathbf{X}$; bold lowercase = column vector, bold uppercase = matrix, plain letter = single number

The regression problem

- Assume the label depends on the features through $y = f(\mathbf{x}) + \varepsilon$: f is a fixed but unknown function and ε is a random error with mean zero, independent of $\mathbf{x}$
- **Supervised learning**: estimate f from the labelled data $\{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^n$, which gives an estimate $\hat{f}$
- **Regression** when the label is quantitative (sales, mpg, balance); classification when it is qualitative (later lectures)
- **Prediction**: compute $\hat{y} = \hat{f}(\mathbf{x})$ for new observations, treating $\hat{f}$ as a black box
- **Inference**: understand how the label changes with each feature (which features matter, how large the effects are, whether the relationship is linear)

Advertising data: questions regression can answer



Advertising data: questions regression can answer (cont.)

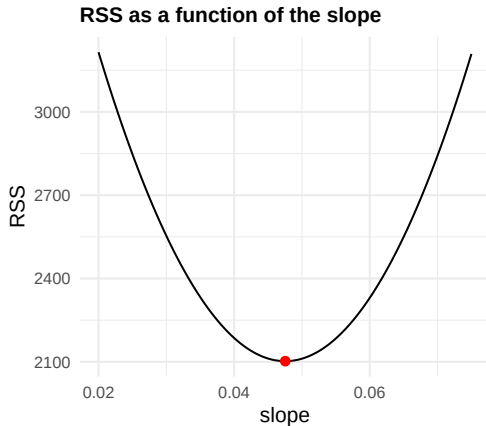
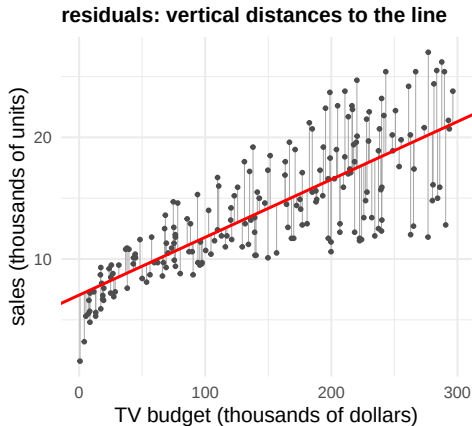
- Is there a relationship between the advertising budgets and sales, and how strong is it?
- Which media are associated with sales, and by how much does each contribute?
- How accurately can we predict sales for given budgets?
- Is the relationship linear, and do the media reinforce each other (synergy)?

Simple Linear Regression

Simple linear regression model

- One feature x and a quantitative label y , related by a straight line plus error: $y = \beta_0 + \beta_1 x + \varepsilon$
- β_0 is the **intercept** (the value of y when $x = 0$), β_1 is the **slope** (the change in y for a one-unit increase in x); together they are the **coefficients**
- Estimate the coefficients from the training data (hat = estimated): $\hat{\beta}_0, \hat{\beta}_1$; predict with $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$
- **Fitted value** of observation i : $\hat{y}^{(i)} = \hat{\beta}_0 + \hat{\beta}_1 x^{(i)}$; **residual**: $e^{(i)} = y^{(i)} - \hat{y}^{(i)}$; residuals are computed from data, the errors $\varepsilon^{(i)}$ are never observed
- Example: $\text{sales} \approx \beta_0 + \beta_1 \times \text{TV}$

Least squares criterion



Least squares criterion (cont.)

- **Residual sum of squares:**

$$\text{RSS} = \sum_{i=1}^n (e^{(i)})^2 = \sum_{i=1}^n (y^{(i)} - \beta_0 - \beta_1 x^{(i)})^2$$

- **Least squares** chooses $\hat{\beta}_0, \hat{\beta}_1$ to minimize RSS: the line with the smallest total squared vertical distance to the points
- Squaring stops positive and negative residuals from cancelling and penalizes large ones more; the red dot marks the minimum (RSS = 2102.5 for TV)

Least squares estimates

Setting the partial derivatives of RSS with respect to β_0 and β_1 to zero gives closed-form estimates, where $\bar{x}$ and $\bar{y}$ are the sample means:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x^{(i)} - \bar{x})(y^{(i)} - \bar{y})}{\sum_{i=1}^n (x^{(i)} - \bar{x})^2}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

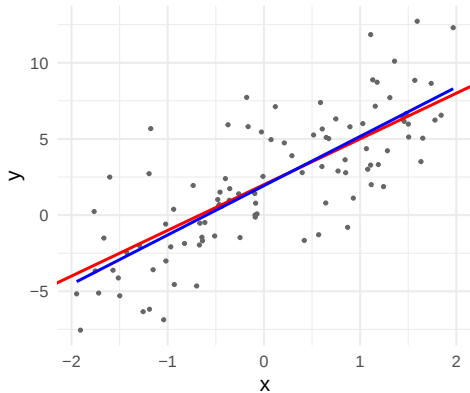
- $\hat{\beta}_1$ is the sample covariance of x and y divided by the sample variance of x
- The fitted line always passes through the point of means $(\bar{x}, \bar{y})$
- Advertising, sales on TV: $\hat{\beta}_0 = 7.03$ and $\hat{\beta}_1 = 0.0475$

Interpreting the coefficients

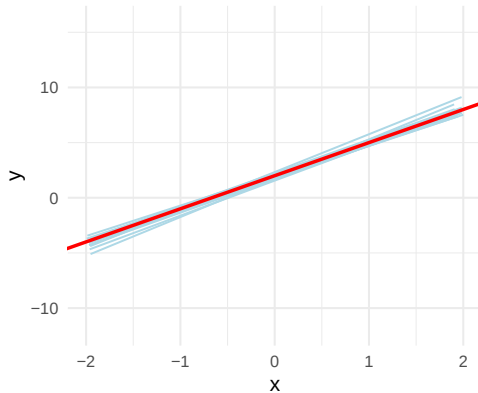
- Fitted line for the Advertising data: $\widehat{\text{sales}} = 7.03 + 0.0475 \times \text{TV}$ (sales in thousands of units, TV budget in thousands of dollars)
- $\hat{\beta}_1 = 0.0475$: an additional \$1,000 of TV budget is associated with about $0.0475 \times 1000 = 47.5$ additional units sold
- $\hat{\beta}_0 = 7.03$: about 7,030 units are predicted when the TV budget is zero; the smallest TV budget in the data is \$700, so the intercept sits at the edge of the data
- The fit describes association in the observed markets; a causal reading needs more than a fitted line, because budgets were not assigned at random

How accurate are the estimates?

one sample: true line (red), fitted line (blue)



ten samples: ten different fitted lines



How accurate are the estimates? (cont.)

- The population regression line (red) is the true, unknown relationship; the least squares line (blue) is computed from one sample, and every new sample gives a different line
- On average the estimates are correct (unbiased), but a single $\hat{\beta}_1$ differs from β_1 ; the **standard error** measures the typical size of this difference

Standard errors of the coefficients

$$\text{SE}(\hat{\beta}_1)^2 = \frac{\sigma^2}{\sum_{i=1}^n (x^{(i)} - \bar{x})^2}, \quad \text{SE}(\hat{\beta}_0)^2 = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{\sum_{i=1}^n (x^{(i)} - \bar{x})^2} \right]$$

- Here $\sigma^2 = \text{Var}(\varepsilon)$ is the variance of the errors; standard errors shrink with more data, less noise, and more spread-out $x^{(i)}$
- σ^2 is unknown and is estimated from the residuals by $\text{RSE}^2 = \text{RSS}/(n-2)$; replacing σ by RSE gives the estimated standard errors reported by software
- These formulas assume the errors are uncorrelated with constant variance σ^2 (checked with residual plots later)
- Advertising, sales on TV: $\text{SE}(\hat{\beta}_0) = 0.4578$ and $\text{SE}(\hat{\beta}_1) = 0.0027$

Confidence intervals and hypothesis tests (recap)

- **95% confidence interval** for β_1 : approximately $\hat{\beta}_1 \pm 2 \cdot \text{SE}(\hat{\beta}_1)$ (exactly, the 0.975 quantile of the t_{n-2} distribution replaces 2)
- **Hypothesis test**: $H_0: \beta_1 = 0$ (no relationship) against $H_a: \beta_1 \neq 0$, using $t = \hat{\beta}_1 / \text{SE}(\hat{\beta}_1)$, which follows a t_{n-2} distribution if H_0 is true
- The **p -value** is the probability of a $|t|$ at least this large if H_0 were true; a small p -value (say below 0.05) is evidence against H_0
- Advertising, sales on TV: 95% confidence interval for β_1 is [0.0422, 0.0528]

	coefficient	std. error	t -statistic	p -value
intercept	7.0326	0.4578	15.36	< 0.0001
TV	0.0475	0.0027	17.67	< 0.0001

Assessing the fit: RSE and R-squared

- **Residual standard error:** $RSE = \sqrt{RSS/(n-2)}$, the typical size of a residual, in the units of y ; Advertising: $RSE = 3.26$, about 3,260 units (23% of the mean sales)
- R^2 : the proportion of the variation in y explained by the regression

$$R^2 = 1 - \frac{RSS}{TSS}, \quad TSS = \sum_{i=1}^n (y^{(i)} - \bar{y})^2$$

- $0 \leq R^2 \leq 1$; Advertising: $R^2 = 1 - 2102.5/5417.2 = 0.612$, so TV explains about 61% of the variance in sales
- In simple regression $R^2 = r^2$, the squared correlation between x and y ($r = 0.782$)
- RSE is in the units of y , R^2 is unit-free; a good R^2 depends on the application

Multiple Linear Regression

Why not several simple regressions?

sales on	$\hat{\beta}_1$	std. error	t -statistic	p -value
TV	0.0475	0.0027	17.67	< 0.0001
radio	0.2025	0.0204	9.92	< 0.0001
newspaper	0.0547	0.0166	3.30	0.0011

- Three separate simple regressions give no single prediction from all three budgets, and each one ignores the other media
- A simple-regression slope also absorbs the effect of every correlated feature that was left out
- Newspaper and radio budgets are correlated ($r = 0.354$): markets that spend more on radio tend to spend more on newspaper, so newspaper may look useful only because it stands in for radio
- Remedy: put all features in one model

Multiple linear regression model

- With p features: $y = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p + \varepsilon$; β_j is the average change in y for a one-unit increase in x_j , *holding all other features fixed*
- For all n observations at once, $\mathbf{y} = \tilde{\mathbf{X}}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$:

$$\underbrace{\begin{bmatrix} y^{(1)} \\ \vdots \\ y^{(n)} \end{bmatrix}}_{\mathbf{y}} = \underbrace{\begin{bmatrix} 1 & x_1^{(1)} & \dots & x_p^{(1)} \\ \vdots & \vdots & & \vdots \\ 1 & x_1^{(n)} & \dots & x_p^{(n)} \end{bmatrix}}_{\tilde{\mathbf{X}}} \underbrace{\begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}}_{\boldsymbol{\beta}} + \underbrace{\begin{bmatrix} \varepsilon^{(1)} \\ \vdots \\ \varepsilon^{(n)} \end{bmatrix}}_{\boldsymbol{\varepsilon}}$$

- $\tilde{\mathbf{X}} = [\mathbf{1} \ \mathbf{X}] \in \mathbb{R}^{n \times (p+1)}$ is the feature matrix with a column of ones for the intercept (the **design matrix**); fitted values $\hat{\mathbf{y}} = \tilde{\mathbf{X}}\hat{\boldsymbol{\beta}}$, residuals $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$

Least squares in matrix form

- The RSS is a function of the coefficient vector:

$$\text{RSS}(\boldsymbol{\beta}) = \sum_{i=1}^n (y^{(i)} - \beta_0 - \beta_1 x_1^{(i)} - \dots - \beta_p x_p^{(i)})^2 = \|\mathbf{y} - \tilde{\mathbf{X}}\boldsymbol{\beta}\|^2$$

- Setting its gradient to zero gives the **normal equations**

$$\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \hat{\boldsymbol{\beta}} = \tilde{\mathbf{X}}^\top \mathbf{y}, \text{ with solution}$$

$$\hat{\boldsymbol{\beta}} = (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^\top \mathbf{y}$$

- The inverse exists when the $p + 1$ columns of $\tilde{\mathbf{X}}$ are linearly independent: this needs $n > p$ and no feature that is an exact linear combination of the others
- Standard errors: $\widehat{\text{Var}}(\hat{\boldsymbol{\beta}}) = \hat{\sigma}^2 (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1}$ with $\hat{\sigma}^2 = \text{RSS}/(n - p - 1)$; $\text{SE}(\hat{\beta}_j)$ is the square root of the diagonal entry for β_j

Interpreting the coefficients

	coefficient	std. error	<i>t</i> - statistic	<i>p</i> - value
intercept	2.9389	0.3119	9.42	< 0.0001
TV	0.0458	0.0014	32.81	< 0.0001
radio	0.1885	0.0086	21.89	< 0.0001
newspaper	-0.0010	0.0059	-0.18	0.860

- Holding radio and newspaper fixed, an extra \$1,000 on TV is associated with about 45.8 more units sold; an extra \$1,000 on radio with about 188.5 more units
- Newspaper is not significant ($p = 0.86$) once TV and radio are in the model, although it was significant on its own ($p = 0.0011$)
- The correlation between newspaper and radio explains it: the simple regression credited newspaper with the effect of radio

Is at least one feature useful? The F-test

- $H_0: \beta_1 = \beta_2 = \dots = \beta_p = 0$ against H_a : at least one $\beta_j \neq 0$

$$F = \frac{(\text{TSS} - \text{RSS})/p}{\text{RSS}/(n - p - 1)}$$

- If H_0 is true, F is close to 1; a value much larger than 1 is evidence against H_0 , with the p -value taken from the $F_{p, n-p-1}$ distribution
- Advertising: $F = 570.3$ on $(3, 196)$ degrees of freedom, $p < 0.0001$: at least one medium is related to sales
- Why not just look at the individual t -tests? With many features, about 5% of the p -values fall below 0.05 by chance even if no feature matters; the F -test accounts for the number of features
- Deciding *which* features to keep is variable selection, deferred to a later lecture

Model fit in multiple regression

model	R^2	RSE
TV	0.612	3.26
radio	0.332	4.27
TV + radio	0.897	1.68
TV + radio + newspaper	0.897	1.69

- $RSE = \sqrt{RSS/(n - p - 1)}$ and $R^2 = 1 - RSS/TSS$; here R^2 is also the squared correlation between $\mathbf{y}$ and $\hat{\mathbf{y}}$
- Adding radio to TV raises R^2 from 0.612 to 0.897 and cuts RSE from 3.26 to 1.68; adding newspaper changes nothing useful
- R^2 never decreases when a feature is added, useful or not, while RSE can increase (the divisor $n - p - 1$ shrinks); adjusted R^2 corrects this (performance measures, later)

Prediction and its uncertainty

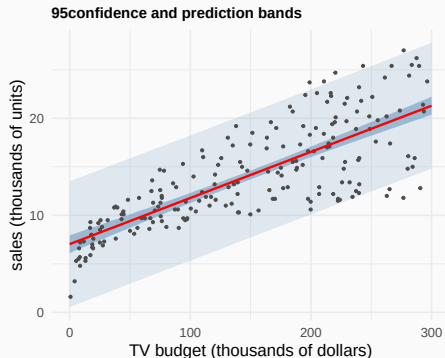
- For a new observation $\mathbf{x}_0 = (x_{0,1}, \dots, x_{0,p})^\top$, with $\tilde{\mathbf{x}}_0 = (1, x_{0,1}, \dots, x_{0,p})^\top$, the prediction is

$$\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_{0,1} + \dots + \hat{\beta}_p x_{0,p} = \tilde{\mathbf{x}}_0^\top \hat{\boldsymbol{\beta}}$$

- Uncertainty from estimating the coefficients (reducible): $\hat{f}$ differs from the true f ; the **confidence interval** covers the *average* response $f(\mathbf{x}_0)$ over many observations with these features
- Model bias: a linear model is an approximation of reality; the intervals assume it is correct (checked with residual plots later)
- Irreducible error: a single response is $y_0 = f(\mathbf{x}_0) + \varepsilon$ even if f were known; the **prediction interval** covers an *individual* response and is always wider than the confidence interval

Confidence and prediction intervals for Advertising

- Sales on TV at TV = 100:
 $\hat{y}_0 = 11.79$, 95% CI [11.27, 12.30],
95% PI [5.34, 18.23]
- The confidence band is narrowest at the mean TV budget and widens towards the edges; the prediction band stays wide because ε dominates
- Adding radio = 20 (TV + radio model): $\hat{y}_0 = 11.26$, CI [10.99, 11.53], PI [7.93, 14.58], narrower because RSE fell from 3.26 to 1.68



Qualitative Predictors and Interactions

Qualitative features with two levels

- Credit data: average credit card balance (dollars) of $n = 400$ cardholders; do women and men differ?
- Encode gender with a **dummy variable**: $x^{(i)} = 1$ if person i is female, 0 if male
- Model $y = \beta_0 + \beta_1 x + \varepsilon$: the mean balance is $\beta_0 + \beta_1$ for women and β_0 for men, so β_0 is the average for men (the **baseline**) and β_1 the average difference between women and men
- Credit fit: $\hat{\beta}_0 = 509.80$ (SE 33.13) and $\hat{\beta}_1 = 19.73$ (SE 46.05, $p = 0.669$): women carry about \$20 more, but the difference is not statistically significant
- Coding male as 1 and female as 0, or as ± 1 , changes what the coefficients mean but not the fitted values

Qualitative features with more than two levels

ethnicity	x_1 (Asian)	x_2 (Caucasian)
African American	0	0
Asian	1	0
Caucasian	0	1

- A feature with k levels needs $k - 1$ dummy variables; the level without one is the baseline; one 0/1/2 variable would impose an order
- $y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \varepsilon$: β_0 is the baseline average, β_1 and β_2 are differences from the baseline
- Credit fit: $\hat{\beta}_0 = 531.00$, $\hat{\beta}_1 = -18.69$ ($p = 0.774$), $\hat{\beta}_2 = -12.50$ ($p = 0.826$)
- The individual p -values depend on the baseline; test the whole feature with the F -test: $F = 0.04$, $p = 0.957$

Interactions: relaxing the additive assumption

- The additive model assumes the effect of TV on sales is the same at every radio budget; spending on both together may work better than the sum of the parts (synergy)
- Add an **interaction term**, a new column of $\tilde{\mathbf{X}}$ with entries $x_1^{(i)}x_2^{(i)}$:
$$y = \beta_0 + \beta_1x_1 + \beta_2x_2 + \beta_3x_1x_2 + \varepsilon = \beta_0 + \beta_1x_1 + (\beta_2 + \beta_3x_1)x_2 + \varepsilon,$$
so the effect of x_2 depends on x_1
- Advertising:
 $\widehat{\text{sales}} = 6.75 + 0.0191 \text{ TV} + 0.0289 \text{ radio} + 0.0011 \text{ TV} \times \text{radio}$; the interaction has $p < 0.0001$, R^2 rises from 0.897 to 0.968 and RSE falls from 1.68 to 0.94
- An extra \$1,000 on radio is associated with about 28.9 more units when $\text{TV} = 0$ and about 137.5 more units when $\text{TV} = \$100,000$
- **Hierarchical principle**: if an interaction is in the model, keep the main effects even when their p -values are large

Interaction with a qualitative feature



Interaction with a qualitative feature (cont.)

- Credit: balance on income and a student dummy; without the income $\times$ student interaction the two groups get parallel lines (same slope, different intercept)
- With it, the slope is $\hat{\beta}_1 = 6.22$ dollars per \$1,000 of income for non-students and $6.22 - 2.00 = 4.22$ for students
- But the interaction estimate -2.00 has SE 1.73 and $p = 0.249$: no strong evidence that the slopes differ

Polynomial Regression

Polynomial regression

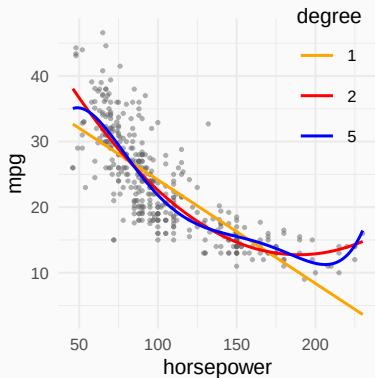
- In the Auto data, mpg is a curved function of horsepower, so a straight line misses systematic structure
- **Polynomial regression** adds powers of the feature:
$$y = \beta_0 + \beta_1 x + \beta_2 x^2 + \dots + \beta_d x^d + \varepsilon$$
- It is still a *linear* regression, linear in the coefficients, with $x, x^2, \dots, x^d$ treated as d features: least squares, standard errors and diagnostics carry over unchanged

$$\tilde{\mathbf{X}} = \begin{bmatrix} 1 & x^{(1)} & (x^{(1)})^2 & \dots & (x^{(1)})^d \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x^{(n)} & (x^{(n)})^2 & \dots & (x^{(n)})^d \end{bmatrix}, \quad \hat{\beta} = (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \tilde{\mathbf{X}}^\top \mathbf{y}$$

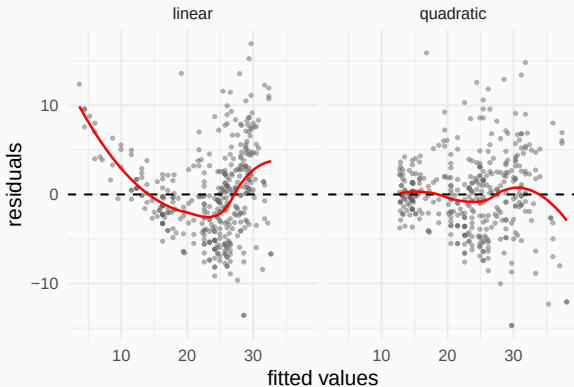
- Quadratic fit ($d = 2$): $\widehat{\text{mpg}} = 56.90 - 0.4662 \text{ hp} + 0.001231 \text{ hp}^2$, both coefficients with $p < 0.0001$

Polynomial fits on the Auto data

fits of degree 1, 2 and 5



residuals vs fitted values



Polynomial fits on the Auto data (cont.)

- R^2 is 0.606 for the linear fit, 0.688 for the quadratic and 0.697 for degree 5: the quadratic term helps a lot, further terms barely
- The linear fit leaves a U-shaped pattern in the residuals, a systematic structure it failed to capture; for the quadratic fit the U-shape is gone
- Higher degrees always improve the fit to the training data, but degree 5 starts to wiggle at the ends: too much flexibility is a risk (bias - variance trade - off, later)

Questions?
